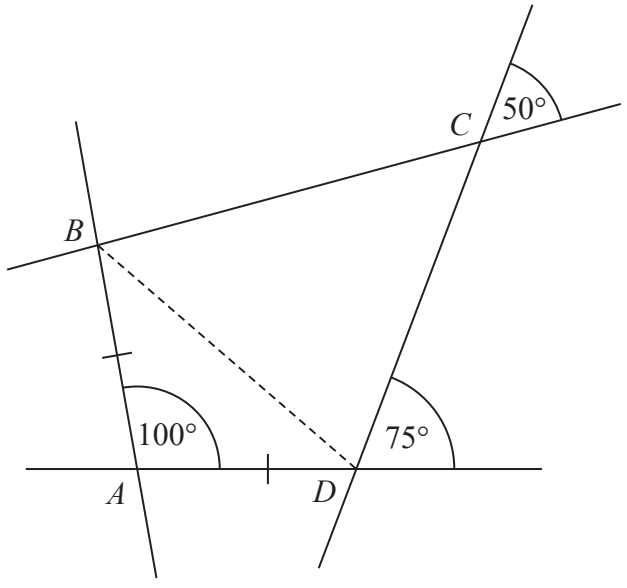
14 The diagram shows quadrilateral *ABCD* with each of its sides extended.



$AB = AD$

Show that *ABCD* is a kite.
Give a reason for each stage of your working.

Angle ADC = 180° – 75° = 105°.

Angles on a straight line add to 180°.

1 mark
Find angle ADC

Angle BCD = 50°: vertically opposite angles are equal.

1 mark
Find angle BCD

Angle ABC = 360° – 100° – 50° – 105° = 105°.

Angles in a quadrilateral add to 360°.

1 mark
Show how to find angle ABC

Join BD. Since AB = AD, triangle ABD is isosceles.

Angles ABD = ADB = (180° – 100°) ÷ 2 = 40°.

So CBD = CDB = 105° – 40° = 65°, giving BC = CD.

AB = AD and BC = CD: two pairs of equal adjacent sides.

Therefore ABCD is a kite.

1 mark
Correct angles and linked geometric reasons

(Total for Question 14 is 4 marks)